

The AI Scientist-v3: Automated Scientific Revision Through Iterative AI Reviewing

Anonymous submission

Abstract

Automated peer reviewers are typically evaluated by the quality of the reviews they produce, but their practical value depends on whether acting on their feedback improves the underlying research. We present AI Scientist-v3, a closed-loop system that begins with an existing rejected paper and iteratively revises its implementation, experiments, and manuscript. The scientist retains its complete revision history, whereas the AI Reviewer independently evaluates only the current manuscript and returns fresh feedback for the next round. We compare three conditions: human-initialized revision, in which written venue reviews guide the first round; AI-initialized revision, in which the AI Reviewer supplies the initial feedback; and autonomous self-revision using the same fixed prompt in every round. Because the AI Reviewer both guides and evaluates the reviewer-driven conditions, we also use Stanford Reviewer as a separate evaluator of the human-initialized trajectories. Reviewer-guided revision improves substantially more than fixed-prompt self-revision, and Stanford Reviewer also assigns higher scores to later versions. AI Scientist-v3 resolves 202 of 237 execution-related weaknesses (85.2%) but only 2 of 28 idea-related weaknesses (7.1%). These results show that iterative reviewer guidance can strengthen the implementation and evidence behind an existing contribution, while concerns about novelty and significance generally require returning to scientific ideation.

Introduction

Scientific publishing is an iterative process rather than a one-time evaluation. Reviewers identify strengths and weaknesses, and authors revise the work before it is considered again. Acting on reviewer feedback, however, often requires substantially more than editing the manuscript: authors may need to reproduce results, run additional experiments, evaluate stronger baselines, analyze new observations, and determine whether the resulting evidence addresses the reviewers' concerns. As language models become increasingly capable of conducting research and reviewing scientific papers, a natural question emerges: Can this process of scientific review and revision itself be automated?

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Recent systems automate important parts of scientific publication: AI scientists can conduct research and prepare manuscripts (Lu et al. 2024; Yamada et al. 2025; Gottweis et al. 2026), while AI reviewers can produce detailed critiques grounded in a submitted paper (Zhu et al. 2025; Goyal et al. 2026; Jiang and Ng 2026). Yet generating a paper or a review does not establish that acting on the review improves the underlying research. In a closed revision loop, the reviewer is not merely an evaluator; it supplies the feedback signal that determines which weaknesses the scientist addresses, much as a grader or reward model directs optimization in reinforcement learning. Even a capable scientist can therefore optimize toward the wrong objective when its reviewer is unreliable. Existing systems provide either a broad experimental action space under fixed human feedback or dynamic AI feedback restricted to paper generation or textual rewriting. What remains unclear is whether an independently evaluated AI reviewer can repeatedly guide code, experiment, and manuscript revisions of the same existing paper without replacing its central contribution.

We study this question using real rejected papers, which already contain an implemented method, experimental code, and written reviews describing the work's shortcomings. We introduce **AI Scientist-v3**, a closed-loop system in which a scientist revises the implementation, executes experiments, analyzes the resulting artifacts, and updates the manuscript. After each revision, an independent **AI Reviewer** evaluates only the current manuscript and returns fresh feedback for the next round. The scientist retains its previous code, results, manuscripts, and feedback so that revisions accumulate; the reviewer retains no history, preventing earlier judgments or scores from biasing its assessment of the current version. The central research contribution remains fixed, allowing us to study revision rather than replacement of the original project.

We evaluate three sources of revision guidance. In the *human-initialized* condition, written venue reviews guide the first round and the AI Reviewer guides subsequent rounds. In the *AI-initialized* condition, the AI Reviewer also supplies the first-round feedback. In *autonomous self-revision*, the scientist instead receives the same fixed self-review prompt in every round. The scientist has the same research capabilities in all three conditions, isolating the effect of the guidance it receives. Because our AI Reviewer both guides and scores

the reviewer-driven conditions, we additionally use Stanford Reviewer to evaluate every saved version from the human-initialized trajectories.

We curate 90 rejected and 10 borderline-accepted ICLR papers spanning multiple research domains. Reviewer-guided revision improves substantially more than autonomous self-revision, while Stanford Reviewer also assigns higher scores to later human-initialized versions. The effect depends strongly on the type of weakness: AI Scientist-v3 resolves 202 of 237 execution-related weaknesses (85.2%), but only 2 of 28 idea-related weaknesses (7.1%). Iterative revision can therefore strengthen the implementation, experimental evidence, and presentation of an existing contribution, but it rarely resolves concerns about the novelty or significance of the contribution itself.

Our contributions are three-fold:

- We introduce AI Scientist-v3, an iterative scientist-reviewer system that revises an existing paper’s implementation, experiments, and manuscript in response to fresh reviews of each version.
- We treat the reviewer as a first-class component: we benchmark its review quality separately, compare its dynamic guidance with fixed-prompt self-revision, and use an external reviewer to evaluate the resulting trajectories.
- We characterize what autonomous revision can and cannot improve: execution-related concerns can often be resolved through systematic revision, whereas novelty-related concerns remain substantially more difficult.

Related Work

AI Scientists and Autonomous Research Autonomous AI systems increasingly cover multiple stages of the scientific research lifecycle, including idea generation, literature review, experimentation, manuscript preparation, and peer review (Lu et al. 2024; Yamada et al. 2025; Gottweis et al. 2026; Baek et al. 2025; Weng et al. 2025). Prior systems already recognize the value of reviewing, but use it in different ways. AI Scientist evaluates a completed manuscript and archives the result for future research generations; AI Scientist-v2 uses VLM critiques to refine experimental and manuscript figures; and CycleResearcher uses CycleReviewer scores to construct preference pairs across training rounds (Lu et al. 2024; Yamada et al. 2025; Weng et al. 2025). These are important precedents, so our claim is not that prior AI scientists omit reviewers. Rather, they do not demonstrate repeated, full-paper peer review driving new code, experiment, and manuscript actions on the same existing rejected paper.

Automated Peer Review Automated reviewers increasingly use structured reasoning, multiple agents, and literature retrieval to produce grounded critiques (Zhu et al. 2025; Goyal et al. 2026; Jiang and Ng 2026). This literature primarily evaluates the review itself. We separately benchmark our reviewer, then ask the distinct downstream question of whether acting on its feedback improves the research.

AI-Assisted Scientific Revision The closest systems each cover one side of our setting. Concurrent APRES repeatedly re-scores and rewrites the same paper, but restricts its action

space to presentation and protects experimental results from modification (Zhao et al. 2026). Concurrent ReviseBench allows repository edits, genuine experiment execution, and manuscript revision, but acts once on a fixed set of human weaknesses rather than obtaining a new review of the revised paper (Luo et al. 2026). AI Scientist-v3 combines the two capabilities: a fresh full-paper review after each version drives another round of executable code, experiment, and manuscript revision on the same rejected work.

Methodology

AI Scientist-v3 consists of two components: a scientist that revises the implementation, experiments, and manuscript, and an AI reviewer that independently reviews each revision and provides feedback for the next iteration. We describe each component below.

AI Scientist-v3 Revising an existing scientific paper differs from generating one from a new research idea. During research generation, the problem formulation, methodology, and experimental design may all evolve as the work develops. Revision instead begins with an established contribution, implementation, and manuscript, and asks which concerns can be addressed without replacing the central research idea. AI Scientist-v3 is designed for this constrained setting.

Given a rejected paper, its source repository, and the current feedback, the scientist first converts each concern into a proposed resolution and checks whether that resolution is supported by the available code and evidence. Its action space includes inspecting the repository and manuscript; searching the literature; editing implementation and configuration files; reproducing existing results; adding baselines, ablations, datasets, or failure analyses; executing those experiments; analyzing the resulting artifacts; and updating figures, tables, prose, and LaTeX. New empirical claims must be grounded in executed results rather than simulated or estimated values. The central contribution is held fixed: if a concern can only be resolved by replacing the research idea, the scientist records it as unresolved rather than presenting a different project as a revision.

Throughout the revision process, the scientist maintains the complete revision history, including previous manuscript versions, reviewer feedback, experimental results, and intermediate analyses. Each revision builds upon the results and decisions from earlier iterations rather than restarting from the original submission. This scientific action space is identical in all three conditions; only the source of guidance differs.

Review Component Our reviewer is inspired by how human reviewers evaluate scientific papers. Rather than assessing every aspect of a paper simultaneously, reviewers gradually build an understanding of the work before forming specific judgments. For example, novelty cannot be assessed without understanding the relevant literature, and experimental results cannot be evaluated without first understanding the claims they are intended to support. Skipping or reordering these steps can lead to unreliable assessments. Our reviewer therefore follows a structured review process in which each

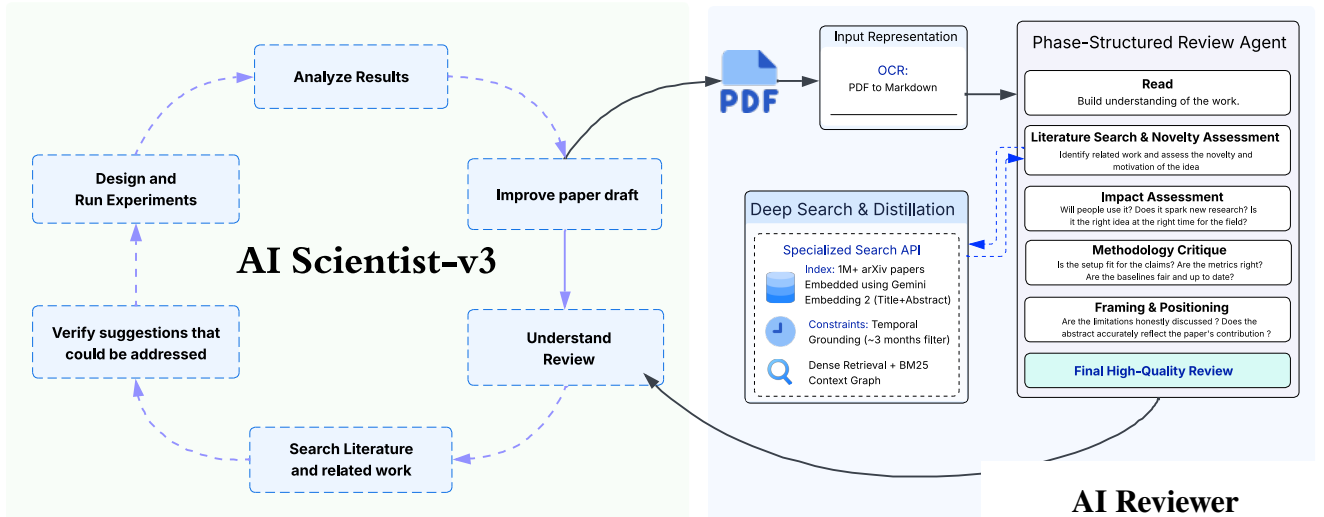


Figure 1: Overview of AI Scientist-v3. The scientist iteratively verifies reviewer suggestions, runs experiments, analyzes results, and updates the manuscript. The review component independently reads each revision, searches the literature, assesses its technical quality and significance, and returns structured feedback for the next round.

stage builds upon the information established by the previous one.

The reviewer first identifies the paper’s contributions, methodology, and evaluation setting before conducting an independent literature search beyond the cited references. To assess novelty, it performs a deep search over an indexed collection of around one million arXiv papers. The search retrieves literature from multiple perspectives, including direct competitors, alternative approaches to the same problem, recent benchmark methods, and work built upon similar baselines. It then evaluates the paper’s novelty, positioning with respect to prior work, the adequacy of its experimental evaluation, and its broader impact before synthesizing these observations into a structured review containing strengths, weaknesses, and suggestions for the next revision.

Unlike the scientist, our reviewer intentionally maintains no memory of previous revisions. Every manuscript version is reviewed independently without access to earlier drafts, reviewer comments, or historical scores. Retaining this history could bias the review by allowing judgments to be influenced by previous evaluations or historical scores rather than the quality of the submitted manuscript. Before producing the final review, the reviewer performs a verification pass to ensure that factual statements and comparisons are supported by either the manuscript or the retrieved literature.

Dynamic Revision Loop Let V_0 and C_0 denote the original manuscript and codebase. Before round t , the scientist receives the preceding artifact (V_{t-1}, C_{t-1}) , its accumulated private history, and guidance G_{t-1} . It then executes actions from the space above to produce updated code, experimental artifacts, and manuscript (C_t, E_t, V_t) . The reviewer sees only V_t (plus literature retrieved during that review) and returns a fresh review R_t without seeing earlier versions, feedback, or scores. In reviewer-guided conditions, $G_t = R_t$ for the next

Table 1: We curate 100 ICLR papers spanning five research domains and retain only papers with publicly available source code that can be reproduced on a single GPU within our VRAM budget.

Domain	N	Decision	
		Rejected	Borderline
NLP	40	37	3
Computer Vision	15	13	2
Graph Learning	15	14	1
Time Series	15	14	1
Reinforcement Learning	15	12	3
Total	100	90	10

round; in autonomous self-revision, G_t remains the same fixed prompt and R_t is retained only for evaluation. Thus “dynamic” refers specifically to regenerating guidance from the newly revised paper, not merely to repeated tool calls within one revision.

Experimental Setup

Data We mine 90 rejected and 10 borderline-accepted ICLR papers with both publicly available reviews and code. To ensure a diverse evaluation set, we sample papers spanning multiple research domains while preferring those with low variance in reviewer scores (Table 1). Finally, we filter papers by computational requirements, retaining only those that can be executed on CPU or a single GPU with 96 GB of VRAM.

Setup We evaluate AI Scientist-v3 under the three revision conditions summarized in Table 2. In the human-initialized



Figure 2: Score trajectories across five scientific domains for the human-initialized condition. Round 0 reports the mean score assigned to the original manuscript by the human venue reviewers. AI Scientist-v3 uses the written venue reviews to produce Round 1. Thereafter, our AI reviewer independently reviews each revision, and its feedback guides the following round.

condition, the original written venue reviews guide the first revision. In the AI-initialized condition, our reviewer instead generates the initial feedback on the original manuscript. In both reviewer-guided conditions, our reviewer independently reviews each subsequent version and its feedback guides the next revision. In the autonomous self-revision condition, the scientist receives no reviewer-generated feedback and uses the same fixed self-review prompt in every round. This comparison isolates the effect of iterative reviewer feedback from the scientist’s ability to revise autonomously. For all conditions, V_0 denotes the original manuscript and V_1 – V_5 denote successive revisions. The scientist retains its complete revision history, whereas our reviewer evaluates each version without access to earlier drafts, feedback, or scores. Each revision run takes approximately 9 hours on a 96 GB VRAM GPU.

Evaluation Our reviewer scores every saved manuscript version without access to earlier drafts, feedback, or scores. Because this reviewer also guides the reviewer-driven conditions, we separately score the human-initialized trajectory with Stanford Reviewer. Stanford Reviewer is used only for external evaluation and its outputs are never shown to the scientist. The score and threshold analyses below use the 19 completed human-initialized trajectories.

Reviewer Evaluation

In a closed revision loop, the reviewer serves both as an evaluator and as the feedback signal that guides the scientist’s next action. Reviewer validity is therefore a prerequisite for interpreting downstream improvements: a stronger scientist does not make the loop reliable if its reviewer consistently rewards the wrong changes. Before using our reviewer’s scores to assess revision, we compare it with existing automatic reviewers using the evaluation setup introduced in DeepReviewBench and ScholarPeer (Zhu et al. 2025; Goyal et al. 2026). We evaluate on papers from 2020–2026, using the ICLR 2024 and ICLR 2025 splits of DeepReviewBench together with the AgentReviewer test set containing ICLR papers from 2020–2023. To evaluate performance on newer research, we additionally include a held-out set of ICLR 2026 submissions. We adopt the extensions to the original

DeepReviewBench protocol, replace the plausibility criterion with significance assessment, and allow the judge to use web search for novelty verification, following ScholarPeer.

We use a search-enabled judge that receives the input paper together with reviews from two anonymized systems, with review order randomized to avoid positional bias. The judge compares the reviews across five dimensions: Technical Accuracy, Constructive Value, Analytical Depth, Significance Assessment, and Overall Judgment, with strong penalties for hallucinated claims and generic feedback.

Results and Discussion

Does Reviewer Guidance Improve Papers? We first evaluate the human-initialized condition, in which the original written venue reviews guide V_1 and fresh feedback from our reviewer guides each later revision. Reviewer scores improve steadily across successive rounds. We compare this condition with autonomous self-revision, in which the scientist receives the same fixed self-review prompt in every round and no reviewer-generated feedback (Figure 3). Although the scientist retains the same literature search, code execution, experiment design, analysis, and manuscript-editing capabilities, removing iterative reviewer feedback substantially reduces improvement. This suggests that the gains arise from the iterative scientist–reviewer process rather than from the scientist alone.

We also evaluate the AI-initialized condition, in which our reviewer generates the initial feedback on V_0 instead of using the venue reviews. This condition consistently outperforms autonomous self-revision and converges to nearly the same final score as the human-initialized condition after five revision rounds.

Figure 2 shows trajectories for the human-initialized condition across different scientific domains. The upward pattern appears in every represented domain rather than being confined to one research area.

External Evaluation. To test whether the improvements in our main human-initialized condition are specific to our reviewer, we separately evaluate every saved version from that condition using Stanford Reviewer. We do not run Stanford Reviewer on the AI-initialized or autonomous self-revision conditions, and its evaluation-only reviews are never pro-

Condition	Guidance used for V_1	Guidance used for V_2-V_5	Role of our reviewer
Human-initialized	Original written venue reviews of V_0	A fresh, independent AI review of the preceding version	Guides V_2-V_5 and scores each version
AI-initialized	A fresh, independent AI review of V_0	A fresh, independent AI review of the preceding version	Guides V_1-V_5 and scores each version
Autonomous self-revision	Fixed self-review prompt	The same fixed self-review prompt in every round	Evaluation only; no feedback is shown to the scientist

Table 2: Revision conditions. Guidance is the feedback shown to the scientist before it produces the indicated version. Evaluation-only reviews and scores are never exposed to the scientist. Stanford Reviewer is applied only to the saved versions from the human-initialized condition.

Category	Baseline	Tech.	Constr.	Analysis	Signif.	Overall
Fine-tuned	CycleReviewer-8B	100/0	100/0	99/1	100/0	–
	CycleReviewer-70B	99.2/0.4	100/0	100/0	99.6/0.2	–
	DeepReviewer-7B	96.5/2.1	99.4/0.4	99.2/0.6	97.8/1.1	–
	DeepReviewer-14B	91.2/3.7	96.0/2.2	94.7/2.6	97.9/1.3	–
Agent w/ Search	Gemini 3.1 Flash	–	–	–	–	–
	Gemini 3.1 Pro	–	–	–	–	–
	DeepSeek V4 Flash	–	–	–	–	–
	DeepSeek V4 Pro	–	–	–	–	–
Multi-Agent Framework w/ Search	Agent Review (Gemini Pro)	–	–	–	–	–
	Agent Review (DeepSeek V4 Flash)	–	–	–	–	–
	AI Scientist v2 (Gemini Pro)	–	–	–	–	–
	AI Scientist v2 (DeepSeek V4 Flash)	–	–	–	–	–
	DeepReviewer-v2	–	–	–	–	–
	Stanford Agent Reviewer	–	–	–	–	–
Ours	AI Reviewer	–	–	–	–	–

Table 3: Side-by-side (SxS) evaluation on DeepReviewBench and ScholarBench. Each entry reports our AI Reviewer’s *Win/Lose* percentages against the baseline in that row.

vided to the scientist. Although Stanford Reviewer assigns different absolute scores, it observes the same upward trajectory across the human-initialized revisions. This agreement reduces the likelihood of optimization to idiosyncrasies of our reviewer, but does not establish human-perceived improvement because both evaluators are AI systems and may share biases.

Score Improvements In the human-initialized condition, our reviewer scores the original papers at 5.37 on average and the best revision for each paper at 6.53, a gain of +1.16 points. Under the same best-revision aggregation, Stanford Reviewer increases from 5.50 to 6.15, a gain of +0.65 points. At V_5 , 13 of 19 papers score at least 6 under our reviewer, compared with 8 at V_0 (Table 4).

What Can Revision Improve? To better understand the limits of autonomous revision, we ask a simple question: **What kinds of reviewer criticisms can revision actually resolve?** To answer this, we first analyse why papers are rejected. We randomly sample 500 rejected ICLR papers and use Gemini 3.1 Pro to classify every reviewer criticism as either an *execution* issue (e.g., missing baselines, weak evidence, unsupported claims, or unclear writing) or an *idea* issue (e.g., insufficient novelty or significance). This analysis is used only to estimate the distribution of rejection reasons

Threshold	V_0	V_5	Δ
Our reviewer			
≥ 5	16/19	18/19	+2
≥ 6	8/19	13/19	+5
≥ 7	2/19	5/19	+3
Stanford Reviewer			
≥ 5.0	17/18	19/19	+2
≥ 5.75	5/18	12/19	+7
≥ 6.0	2/18	8/19	+6

Table 4: Papers crossing score thresholds at V_0 and V_5 in the human-initialized condition. Our reviewer uses whole numbers; Stanford’s scale is continuous. One original paper lacks a Stanford score, producing the 18-paper denominator in that column.

and is separate from our evaluation set. Execution issues account for 48% of rejections, idea-related issues for 35%, and the remaining 17% involve both.

Having identified these two broad categories, we next ask which of them can actually be resolved through iterative revision. On our evaluation set, AI Scientist-v3 resolves 202

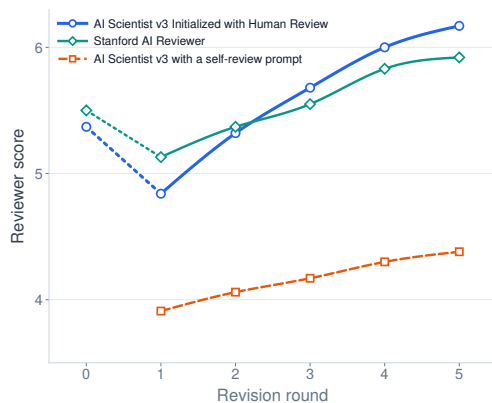


Figure 3: Under our reviewer, the human-initialized condition improves substantially more than autonomous self-revision with a fixed prompt. Stanford Reviewer, applied only to the human-initialized trajectory, also assigns higher scores to later revisions.

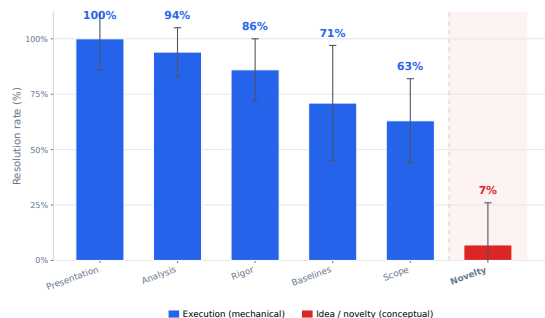


Figure 4: Resolution rate of AI-suggested weaknesses by AI Scientist-v3. Most empirical weaknesses are resolved, whereas only 2 of 28 idea-related weaknesses (7.1%) are addressed, suggesting that iterative revision is more effective at improving research execution than altering the core scientific idea.

of 237 execution weaknesses (85.2%), but only 2 of 28 idea weaknesses (7.1%). As shown in Figure 4, revision consistently succeeds on suggestions that require stronger evidence via additional experiments, baselines, ablations, or clearer presentation, but rarely changes reviewer judgements about novelty or significance.

This difference persists across revision rounds. Our reviewer retains novelty-related objections until the final round, often citing prior work to justify them. Once the contribution itself is judged to lack novelty, additional evidence seldom changes the outcome. Our results therefore suggest that autonomous revision is well suited to strengthening the evidence behind an idea, but improving the idea itself requires returning to the ideation stage rather than continuing the revision loop.

Conclusion

AI Scientist-v3 demonstrates that iterative reviewer guidance can improve rejected papers, particularly by strengthening their implementation, experimental evidence, and presentation. Both human- and AI-initialized revision outperform autonomous self-revision with a fixed prompt by the final round. In the human-initialized condition, Stanford Reviewer also assigns higher scores to later revisions than to the original manuscript, reducing—but not eliminating—the possibility that the observed gains are specific to our reviewer. These improvements are concentrated in execution-related weaknesses; concerns about novelty and significance are rarely resolved through revision alone. Our results therefore position autonomous revision as a tool for strengthening the evidence and execution behind existing ideas, rather than as a replacement for scientific ideation.

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